YUXUAN HOU

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Education

Southern University of Science and Technology (SUSTech) — Expected Jun 2028

B.Eng. in Computer Science and Technology — Turing Class (Honors Program) — *Shenzhen, China*

Academic-year GPA (2025–2026): 3.86/4.00 | **Class President**, 2024 CSE Turing Class

Honors Coursework: Discrete Mathematics; Data Structures and Algorithm Analysis; Algorithm Design and Analysis; Principles of Database Systems; Artificial Intelligence.

Other Completed Coursework: Computer Organization; Computer Vision; Natural Language Processing; Digital Logic; Introduction to Mathematical Logic; Introduction to Computer Programming; Computational Ethics; Group Projects I.

In Progress: Operating Systems (Honors); Compiler Principles; Computer Networks; Advanced Computer Programming; Probability and Statistics for Engineering.

Research Affiliation

Lab: **Advanced Intelligent Mobile Systems (AIMS) Lab**, SUSTech

Advisor: **Jin Zhang**, Associate Professor, Department of Computer Science and Engineering

Accepted Publication

Yuxuan Hou, Jiao Li, Linshan Jiang, and Jin Zhang. *Oto-Meal: Earable Sensing with PPG and IMU for Personalized Meal Awareness*.
WellComp 2026 (UbiComp/ISWC 2026 Workshop) — Computing for Well-being.
Accepted; forthcoming in *UbiComp Companion '26*. DOI: 10.1145/3798063.3841767. arXiv:2608.16678.

Research Experience

Oto-Meal: Earable Sensing with PPG and IMU for Personalized Meal Awareness — Sep 2025–Present

Project Lead & Primary Developer — *Wearable sensing*

First-author paper accepted to **WellComp 2026 (UbiComp/ISWC 2026 Workshop)**.

Project Lead — 2026 National Undergraduate Innovation and Entrepreneurship Training Program (**National Level**; CNY 25,000)

Project Lead — 2026 SUSTech "Climbing Program" Project (CNY 15,000; completed)

- Built an audio- and image-free earable prototype using photoplethysmography (PPG) and inertial sensing; evaluated a pooled neural classifier separately from within-user memory matching on **538 recordings and 7,028 windows from 7 participants**.
- Continue follow-up research on data-efficient recognition and transfer learning; completed a **32-case pilot** on cross-user evaluation.

Neuromorphic Sensing for Edge Devices — Ongoing

Method Development & Device Evaluation — *In preparation*

- Develop spiking neural network (SNN) approaches for wireless and wearable sensing, studying accuracy and resource trade-offs across **ten datasets**.
- Evaluate on **Jetson AGX Orin** across three datasets and three seeds, auditing **81 inference conditions** for accuracy, latency, throughput and power measurements.

LADS: Lease-Aware Delivery Scheduling of Tool Results for Language Model Agents — 2026

Systems Research & Prototyping — *Submitted*

- Coordinate tool-result delivery and cache reuse for concurrent language-model agents. Submitted to **WWW 2027**; arXiv submitted (awaiting announcement).
- Reduce mean complete-workflow latency by **44.6%** on controlled bursts (5 seeds) and **42.4%** on Web-derived burst replay (3 seeds), versus ReAct with first-in, first-out delivery, using the same model and agent policy.

Research Experience (continued)

3D Acoustic Field Reconstruction and Adaptive Acoustic Modeling **May 2026–Present**

Acoustic Modeling & Simulation *In preparation*

- Study acoustic scene reconstruction from sparse measurements and measurement-geometry calibration; reproduce prior work, develop simulation and evaluation workflows, and assess reconstruction and calibration behavior.

Related computer vision project: Team Leader, four-person Object-Level 3D Gaussian Splatting project (Spring 2026); built reconstruction, segmentation and editing pipeline, registering 213/213 frames with 0.71-pixel reprojection error.

RydbergTrack (Spring semester 2026; Team Leader, three-person project): built a reproducible MATLAB pipeline for 150 million samples; separated 3 static and 12 motion recordings within the project dataset.

Technical Skills

Programming: Python, C/C++, MATLAB, Java, Verilog
ML & Scientific Computing: PyTorch, scikit-learn, NumPy, SciPy; spiking neural networks, signal processing, numerical optimization
Systems & Tools: CUDA, COLMAP, Vivado, Git, Linux, LaTeX; inference scheduling, reproducible evaluation, automated testing

Selected Honors & Awards

2025 **National First Prize**, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)
2025 **Silver Medal (Rank 4)**, International Collegiate Programming Contest (ICPC)
2025 **Silver Medal**, China Collegiate Programming Contest (CCPC)
2025 **Gold Medal**, Guangdong Collegiate Programming Contest
2026 **International First Prize (1st Place)**, Application Development Division Finals, 1st Hong Kong International AIGC Cultural Digital Content Creation Competition
2025 Provincial First Prize, 16th Lanqiao Cup, Guangdong; C/C++ Programming, University Group A
2024/25 SUSTech University Scholarship; Zhiren College Motto Scholarship “Rixin”; SUSTech Outstanding Student Leader
2026 Undergraduate Talent Development Excellence Program Grant (2024 Cohort; CNY 6,000)
2025/26 SUSTech Outstanding Communist Youth League Cadre
2026 Second Prize, CCF Service Computing Innovation Competition / Sangfor CoStrict AI-Native Innovation Competition; *GeoAgent* (team project)

Earlier informatics honors: Provincial Rank 9, National Olympiad in Informatics (NOI) Joint Provincial Selection; First Prizes in the NOI Spring Test and CCF CSP-S (senior-level certification).

Leadership & Service

President, Student Union | Chair, Student Development Committee: lead organizational planning, team coordination, recruitment, student activities, member development, leadership handovers, and operating procedures.

Head, Zhiren College Peer Mentor Team: lead 15 mentors serving 231 incoming students across 5 classes and outreach covering 25 provinces; organize onboarding, orientation, and ongoing support.

Collective honors: SUSTech Outstanding Student Union; First Prize, Zhiren College Outstanding Student Organization (2025–2026).

Volunteer service: 87.17 verified hours, including campus events and accessibility initiatives.

Other roles: Turing Class President; 3D Manufacturing Club Vice President; competitive programming, supercomputing, rowing, emcee, and exhibition hall docent teams.